

TUTORIAL -1 SDLC & STLC PHASES

Aim: - Understand SDLC and STLC Phases for a Real-World Software Project and Identify Testing Activities Performed During Each Development Stage.

Description :

Software Development Life Cycle (SDLC) is a structured process used to design, develop, test, deploy, and maintain software applications. Software Testing Life Cycle (STLC) is a sequence of testing activities performed to ensure the quality of software.

In this tutorial, a Student Management System is considered as a sample project to understand the relationship between SDLC and STLC phases. Different testing activities performed during each development stage are identified and documented.

Problem Statement :

A college requires a Student Management System to maintain student records, course registration, attendance, examination details, and result generation. The software should allow administrators to manage student information efficiently while ensuring data accuracy and security.

The testing team must identify testing activities performed during every SDLC phase and prepare test cases for the Login Module.

Implementation Details :

Parameter	Details
Project Name	Student Management System
Application/Software	Web Application
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE/Tool Used	Visual Studio Code
Input/Test Data	Username: admin, Password: Admin@123

Source Code :

```
username = input("Enter Username: ")
```

```
password = input("Enter Password: ")
```

```
if username == "admin" and password == "Admin@123":
```

```
    print("Login Successful")
```

```
else:
```

```
    print("Invalid Username or Password")
```

Test Case Detailed :

Field	Details
Test Case ID	TC_LOGIN_001
Requirement ID	REQ_LOGIN_001
Module Name	Login
Feature Under Test	User Authentication
Test Scenario	Verify valid login credentials
Test Objective	Validate successful login functionality
Test Type (Unit/Integration/System/UAT)	System Testing
Test Technique (Black Box/White Box)	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Login page should be accessible
Postconditions	User redirected to Dashboard
Test Environment	Windows 11, Python 3.12
Browser/OS (if applicable)	Google Chrome / Windows 11
Test Data	admin / Admin@123
Expected Result	User should successfully login
Actual Result	Login Successful
Status (Pass/Fail)	Pass
Executed By	Student
Execution Date	

Test Case Execution :

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
Open Login Page	URL	Login page displayed	Login page displayed	Pass	Open Login Page
Enter Username	admin	Username accepted	Username accepted	Pass	Enter Username
Enter Password	Admin@123	Password accepted	Password accepted	Pass	Enter Password
Click Login Button	Mouse Click	Dashboard displayed	Dashboard displayed	Pass	Click Login Button
Verify Dashboard	-	User logged in successfully	User logged in successfully	Pass	Verify Dashboard
Open Login Page	URL	Login page displayed	Login page displayed	Pass	Open Login Page
Enter Username	admin	Username accepted	Username accepted	Pass	Enter Username

Testing Activities Performed :

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Analysis	Analyze software requirements	SRS, RTM
2	System Design	Test Planning	Prepare Test Plan and Test Strategy	Test Plan
3	Development	Test Case Development	Design Test Cases	Test Case Document
4	Testing	Test Execution	Execute Test Cases	Test Execution Report

5	Deployment	Test Closure	Verify deployed application	Test Summary Report
6	Maintenance	Regression Testing	Validate software after updates	Defect Report

Observation :

Sr. No.	Observation	Remarks
1	SDLC defines software development phases.	Successfully understood
2	STLC focuses only on software testing.	Correct
3	Every SDLC phase has related testing activities.	Verified
4	Login module passed all test cases.	No defects found
5	Documentation is important throughout the lifecycle.	Completed

Output :

Successfully studied the **Software Development Life Cycle (SDLC)** and **Software Testing Life Cycle (STLC)**.

Identified testing activities corresponding to each SDLC phase.

Prepared and executed test cases for the Login Module.

Verified that the application functions correctly using valid test data.

Understood the relationship between software development and software testing processes